GY462—Real Estate (RE) Finance

Lecture 2

1

Discounted Cash Flow Valuation in RE Context – *Revision*

- **The method**
  1) **Forecast expected future cash flows**
  2) **Ascertain required total returns (discount factor or proxy for 'risk')**
  3) **Discount cash flows to present value at the required rate of return**

⇨ **Find value such that expected return equals required return** (NPV=0)

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What Should the Discount Rate Reflect?

- **Should reflect risk of project**
  - $r = r_f + RP$
  - r_f: **Risk free rate (Treasury bill yield; time value of money)**
  - *RP:* **Risk premium**
- **Risk premium does not reflect risk in absolute term, but contribution to risk of overall portfolio**

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Sources of Discount Rate?

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Discount Rate and Risk Premium for Longer Term Projections

- Typical 'real world' commercial property analysis uses <u>10 year projection of CFs</u>
- Can we assume fixed discount rate like in previous example?
  - What if current lease expires in year 6?
  - Initial lease has lower risk
    - Existing contract → known cash flows
    - Risk limited to tenant default risk
  - Economic situation when lease is expiring?

Example with Lease Renewal

- Consider 10 year projection
- First 6 years: existing lease
- Next 4 years: expect new lease (with uncertainty)
- Expected sale price in year 10: 20
- Required rate of return:
  - 8% first 6 years
  - 15% next four years (more risky)

⇨ DCF valuation in year 2000?

Year	CF
2001	1.0
2002	1.0
2003	1.0
2004	1.5
2005	1.5
2006	1.5
2007	2
2008	2
2009	2
2010	2

Example with Lease Renewal – *Continued*

Solution:

$FV_{t=10} = 20$

$i_t = 0.08 \;\; \forall \;\; t=1,...,6 \;\; ; \;\; i_t = 0.15 \;\; \forall \;\; t=7,...,10$

$n = 10$

$CV_t = 1 \;\; \forall \; t = 1,...,3 \;\; ; \;\; CV_t = 1.5 \;\; \forall \; t = 4,...,6 \;\; ;$

$CV_t = 2 \;\; \forall \; t = 7,...,10$

$$PV_{t=0} = \sum_{t=1}^{3} \frac{1}{(1.08)^t} + \sum_{t=4}^{6} \frac{1.5}{(1.08)^t} + \sum_{t=7}^{10} \frac{2}{(1.15)^t} + \frac{20}{(1.15)^{10}}$$

$= 13.058$

Example with Lease Renewal – *Continued*

⇨ **Is this really the 'only' solution?**

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Intra/Inter-Lease Discount Rate

- **Distinguish between CFs covered by signed lease and CFs/sales price expected but not guaranteed**
  - ***Intra-lease* (within) discount rate: Reflects tenant default risk**
  - ***Inter-lease* (between) discount rate: Reflects rental market risk/uncertainty about sales price**

⇨ **What is the correct DCF-value in this case?**

Solution – *Step by Step*

- **Risk components?**
  - **Risk associated with initial lease**
    - **Limited to tenant default risk**
    - **CFs are known (existing contract)**
  - **Risk associated with second lease**
    - **Economic situation when lease is expiring?**
    - **Once 2nd lease is signed: tenant default risk?**

Solution

Step 1: Expected PV in t=2000 of first lease (6 yrs CF)

$CV_t = 1 \;\forall\; t = 2001,\ldots,2003 \;;\; CV_t = 1.5 \;\forall\; t = 2004,\ldots,2006$

$$PV_{t=2000} = \sum_{t=2001}^{2003} \frac{1}{(1.08)^{t-2000}} + \sum_{t=2004}^{2006} \frac{1.5}{(1.08)^{t-2000}}$$

$= 2.577 + 3.069 = 5.646$

Intra-lease discount rate

Solution – *Continued*

Step 2: Expected PV in t=2006 of second lease (4 yrs CF)

(Consider 2nd lease as if it were 2006 and lease with expected conditions is signed...)

$CV_t = 2 \;\;\forall\;\; t = 2007,\ldots,2010$

$$PV_{t=2006} = \sum_{t=2007}^{2010} \frac{2}{(1.08)^{t-2006}}$$

$= \underline{6.624}$ (value in 2006 not 2000!)

Intra-lease discount rate

How much is 6.624 in 2000?

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Solution – *Continued*

Step 3: Expected PV of second lease in t=2000

FV of second lease: 6.624 (in t=2006)

PV of second lease in t=2000: ?

Discount rate? ⇒ 15% (Value more uncertain)

$$PV_{t=2000} = \frac{6.624}{(1.15)^{2006-2000}}$$

$= \underline{2.864}$ (value in 2000!)

Inter-lease discount rate

Solution – *Continued*

Step 4: Expected PV of property sale in 2010?

FV of property: 20 (in t=2010)

PV of property in t=2000: ?

Discount rate? ⇒ 15% (Interlease discount rate!)

$$PV_{t=2000} = \frac{20}{(1.15)^{2010-2000}}$$

$= \underline{4.944}$ (value in 2000!)

Solution – *Continued*

Step 5: DCF value of property in 2000?

PV of 1st lease in t=2000: 5.646

PV of 2nd lease in t=2000: 2.864

PV of sale in year 10 in t=2000: 4.944

DCF Valuation in 2000: 5.646+2.864+4.944=13.454

Intra/Inter-Lease Discount Rate – *Summary*

⇨ **Alternative valuation to better reflect risk:**

Intra-lease discount rate: risk *during lease-term* reflecting risk that tenant might default on contractual lease obligation

13.453 =

$$PV_{t=0} = \sum_{t=1}^{3} \frac{1}{(1.08)^t} + \sum_{t=4}^{6} \frac{1.5}{(1.08)^t} + \left(\frac{1}{(1.15)^6}\right) \sum_{t=1}^{4} \frac{2}{(1.08)^t} + \frac{20}{(1.15)^{10}}$$

Inter-lease discount rate: higher risk reflecting uncertainty about future lease contract (rental market risk) and uncertainty about sales price

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'Focus point 2'

Intra- vs. inter- discount factor;
tenancy vs. lease risk:
Are these terms clear? How are they related?

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Conclusion?

⇨ **Given similar estimates of**

- CFs and sales price at end of projection
- Tenant default risk and discount rate associated with uncertain future economic conditions

- **Sophisticated investor can earn a fortune**
  - In example: £396,000!
  - This is the difference between simple price (13.058) and the inter-intra price (13.454)

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In Practice – *Some caveats*

- **Complicating factors**
  - **Variety of leases (and lease-expiry dates) within same property**
  - **Cash flows not entirely fixed: uncertainty on operating expenses and other determinants of cash flow**

➔ **Mechanical, but sometimes complex, 'matching' exercise!**

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Alternative valuation methods?

- **Appraisal based on income produced by property not unique**

- ***Sales comparison* approach**
  - **Analyze data of recently sold highly comparable properties**
  - **Adjust for differences**
  - **Rationale? No-one would pay more for a property than other recent investors**

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Alternative valuation methods? *Continued*

- ***Construction cost*** **approach**
  - Analyze construction costs and cost of buying land
  - For existing buildings: discount physical, functional and external obsolescence
  - Rationale? No-one would pay more for a property than it would cost to build it
- Various methods should give similar values if well implemented

Contents — *Chapter 1*

1. Investment Analysis
   - Valuation of Income Producing Properties
     - DCF Method in Real Estate Context
     - Choice of Discount Rate
     - Shortcut Methods: Cap Rate, GIM and Equivalent Yield
   - Pro Forma Basics
   - Investment Decision
     - Measures of Investment Performance
     - NPV and IRR Investment Rules
   - Risk Assessment Methods

Other Short(er)cut Valuation Methods

- Short cut <u>valuation</u> methods

  1. Direct capitalization (cap rate)
  2. Gross income multiplier (GIM)
  3. Equivalent yield

Shortcut: Direct Capitalization (Cap Rate)

- Skip multi-year CF forecast
- Compute <u>market</u> observed cap rate as
  - Cap rate = Average (NOI ÷ V)
- Compute value of <u>our</u> property as
  - V = NOI ÷ Cap rate
- Requires market observation for similar properties
  - To determine what cap rate to apply to our property
  - Plus an estimate on initial NOI

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Terminology

- Assume investment lasting for ten years…
- 'Going in' cap rate
  = average (year-1 NOI ÷ V)
- 'Going out' (or terminal) cap rate
  = average (year-11 NOI ÷ V)
- 'Stabilised' cap rate
  = average (stabilised NOI ÷ V)

Cap Rate — *Example*

- Suppose 'going in' cap rate for typical single tenant office building in particular market is 7.5%
- The initial NOI is 1
- Value of property = 1 ÷ 0.075
  = 13.333

What are Typical Cap Rates? No Simple Answer…

- Cap rates <u>vary over time</u> and from country to country (macro factors)
- Cap rates measure <u>how risky particular types of property are</u>
- Examples
  - What happened to cap rates in New Orleans after Hurricane Katrina?
  - What happened to cap rates of office buildings relative to beach hotel buildings in Thailand after the tsunami?

The Mechanics of Cap Rates

- Consider two properties with similar initial NOI. Worth how much?
  - When cap rate is high the property is worth little
  - When cap rate is low the property is worth more
- What does this mean? If similar NOI but different cap rates…
  - *Low CR:* valuable property with better appreciation prospects, located in growing areas, less depreciation
  - *High CR:* Less valuable property attached to more risk and uncertainty

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Cap Rates for Different Types of *Stabilised Properties* in the US

Description	*Cap Rate (April 2010)*
Office: *Downtown office, prime towers, well designed, top of market, big building, quality tenants, smooth lease expiration*	**7 - 8**
Office: *Suburban, good, smaller, fewer tenants, not as well located, not quite as liquid*	**7.5 - 9**
Office: *Second tier city, suburban, small, two tenants, med. quality credit tenants, less liquid, lumpy lease expiration*	**8.5 – 12**
Garden Apartments (A quality): *Good location, recent design*	**7 - 8**
Garden Apartments (B quality)	**8 - 9.5**
Garden Apartments (C quality)	**9 - 12**

Source: Linneman (2013)

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Cap Rates for Different Types of Stabilised Properties in the US *(Cont.)*

Description	*Cap Rate (April 2010)*
Shopping Centres (A): *Good supermarket, anchored tenant with known stores in competing market (no fear of sudden entry and decline of customers)*	**7.5 – 8.5**
Shopping Centres (B): *Pretty good strip centre*	**8.5 - 11**
Shopping Centres (C): *Empty space, low quality tenants, not well located*	**10 - 14**
Warehouse: *Great location, near major highway or exit ramp*	**8 - 9.5**
Warehouse: *Poor location, no easy access to highway*	**9 - 12**
Hotels: *Incredibly difficult to value, huge variation in cap rates*	**n/a**

Source: Linneman (2013)

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'Focus point 3'

**High cap-rate vs. low-cap rate: is the logic clear?
Location, structures, tenants (Lec.1):
can you see it here?**

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Problem of Single Year Ratio Valuation Methods

- **Annual cash flow projections for two identical risk buildings A and B**

Year	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
A	1	1	1	1	1	1	1	1	1	9.333
B	1	1.02	1.04	1.061	1.082	1.104	1.126	1.149	1.172	13.385

- **Same cap rate assumption leads to misleading results!**
- **How much would you be willing to pay for buildings A and B assuming a required return of 12%?**

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Cap Rate Adjustments — *Example*

- Use DCF method
  - $V_A = 8.33$; $V_B = 10$
- What would be the correct corresponding cap rate?
  - Cap rate A: 12% ; Cap rate B: 10%
- Difference of 2% equals growth rate difference ⇨ coincidence?

Cap Rate Adjustments — *Continued*

- Properties A and B are both a special case of the constant multiple & constant growth perpetuity
  - CF stable or grows at constant rate
  - Asset value always a constant multiple of current CF
- For this special case: $V = \frac{CF_1}{(r-g)}$
  - With *cap rate* = required/expected IRR r – constant CF growth rate g
  - Cap rate of building B: 10% = 12% - 2%

    A: 12% = 12% - 0%

Cap Rate Adjustments — *Continued*

- Real world buildings often come close to these regularity assumptions
  - Cap rate approximation ('Gordon Model'):

    $Cap\ Rate = (r-g)$
  - Are the algebra and logic clear?

Gordon Model vs. DCF Analysis? *Discuss*

- Gordon Model
  - 'Quick and dirty' calculation
  - Generates often almost same answer as full DCF analysis
- Critical Assumptions
  - NOI growth is fairly steady (stabilized)
  - NOI growth is small relative to discount rate
- If property does not meet stabilized criteria, need to do full DCF analysis

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Break… ☺

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Equivalent Yield — *Brief overview*

- **Most commonly used valuation method in Britain**
  - **Only brief discussion**
  - **Very close to cap-rate**
- **Consists of two parts**
  - **Current income up to next rent review**
  - **At review: current estimate of rental value**

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Equivalent Yield

- **Property value V is estimated as:**

$$V = \frac{I}{y} + \frac{R - I}{y(1+y)^n}$$

where

R…current rental value

I…passing income

y…equivalent yield

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Where We Are…

1. **Investment Analysis**
   - Valuation of Income Producing Properties
     - DCF Method in Real Estate Context
     - Shortcut Methods: Cap Rate, GIM and Equivalent Yield
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   - **Pro Forma Basics**
   - Investment Decision
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Remember…

- DCF valuation involves two major analytical steps
  1) Forecast future expected net cash flows from property
  2) Determine and apply appropriate opportunity cost of capital as a discount rate

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Remember…

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Proforma

- A multi-year cash flow forecast
- Shown to lenders and outside investors
- 2 types of cash flows
  - Operating
  - Reversion

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Operating versus Reversion CF

- Operating CFs
  - CFs that result from normal operation of property
  - Accrue to owner regularly throughout holding period
- Reversion CFs
  - Occur only at time of sale and occur only due to sale of all or portion of asset

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Operating CF Proforma —*Typical Line Items*

- **Operating (all years)**

Potential Gross Income (Rent*SF)	=	PGI
– Vacancy allowance (=Vac Rate*PGI)	=	– V
+ Other Income	=	+ OI
– Operating expenses	=	– OE
= Net Operating Income	=	NOI
– Capital Improvement Expenditures	=	– CI
(O) Property Before-Tax Cash Flow	=	PBTCF

Operating CF Proforma —*Top Line*

- **Potential gross income (PGI)**
  - ***'Rent Roll'***
  - **Other potential sources of property revenue**

'Focus point 4'

'Rent roll': what to consider?
Other sources of income: which ones?

Vacancy Allowance

- **For each space/lease: forecast probability of renewal and expected vacant period if not renewed**
- **Vacancy**
  - **= vacant months / (vacant + rented months) in typical cycle**
  - **Look at typical vacancy rate in rental market or history of subject property**

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Operating Expenses

- **Fixed**
  - **Property taxes**
  - **Hazard insurance**
  - **Property Security**
  - **Management**
- **Variable**
  - **Maintenance and repairs**
  - **Utilities not paid by tenants**
- ***Does not include:*** **Income taxes, depreciation expenditures**
- ***Includes:*** **Management expenses, even if self-managed**

Capital Improvement Expenditures

- **Leasing costs (one-shot)**
  - **Tenant improvement expenditures (TIs)**
    - Customized physical improvements
  - **Leasing commissions to brokers**
- **Property improvements**
  - **Major repairs**
  - **Replacement of major equipment**
  - **Major remodeling of building, grounds, fixtures**
  - **Expansion of rentable area**

'Focus point 5'

Capital Improvements:
expenditures or investments?
Can you think of some relevant examples?

Reversion CF Proforma —*Typical Line Items*

- **Reversion**
  (last year & years of partial sales)

Property value at time of sale	**=**	**PGI**
– Selling Expenses	**=**	**– SE**
(R) Property Before-tax Cash Flow	**=**	**PBTCF**

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Resale Value
– *How to Determine?*

- **Consider typical 10 year DCF valuation**
  - **Projected reversion typically > 1/3 of PV**
- **Widespread resale forecasting method**
  - **Divide year 11 NOI by 'going out' cap rate**
- **Going out cap rate relative to going in cap rate?**
  - **Usually going out cap ≥ going in cap**
  - **Why?**

'Focus point 6'

Going out cap ≥ going in cap?!
How do I make money (i.e., sell at a premium)?

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Measures of Performance
— *Overview*

- **Measures of investment performance using ratios**
  - **Price per square foot**
  - **Cap rate**
  - **Equity dividend rate**
  - **Debt coverage ratio**

  Short-cut methods *– in brief*

- **Measures of investment performance based on CF projections**
  - **NPV**
  - **IRR**

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NPV Investment Rule

- **Maximize NPV across all mutually exclusive alternatives with NPV≥0**
- **Zero NPV deals are ok and expected!**

NPV(Buyer) = Market Value – Price
NPV(Seller) = Price – Market Value

⇨ NPV(Buyer) = – NPV(Seller)

Since deal only if NPV(Buyer) = NPV(Seller)
Thus NPV(Buyer)=NPV(Seller)=0

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IRR Investment Rule

- **Question: What discount rate will cause expected future cash flows to be worth the given price?**
- **Rule:**
  1. **Maximize difference between project's expected IRR and required return (i.e. benchmark rate + RP)**
  2. **Never do deal with expected IRR less than required return ("hurdle rate")**

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IRR Computations – *Details*

- **Consider 10 year projection**
- **First 6 years: existing lease (step-up)**
- **Next 4 years: expect new lease, with uncertainty but higher value**
- **Expected sale price in year 10: 20**
- **I am being asked to pay 13.453 for this property**

⇨ **IRR of investment?**

Year	CF
2001	1.0
2002	1.0
2003	1.0
2004	1.5
2005	1.5
2006	1.5
2007	2
2008	2
2009	2
2010	2

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How to compute IRR with calculator

IRR: Discount rate such that PV of future cash inflows = initial cash outflow

Financial Calculator Solution (TI BA II PLUS)

CF	(Select cash flow worksheet)		
CF0=	–13.45	ENTER	(Enter initial cash payment, PV)
C01=	1	ENTER	(enter 1st CF)
F01=	3	ENTER	(enter n. of periods of 1st CF)
C02 =	1.50	ENTER	(enter 2nd CF)
F02=	3	ENTER	(enter n. of periods of 2nd CF)
C03=	2	ENTER	(enter 3rd cash flow)
F03=	3	ENTER	(enter n. of periods of 3rd CF)
C04=	22	ENTER	(enter last period CF+sale value)
F04=	1	ENTER	(enter n. of periods of last CF)
IRR CPT	=13.13		(computes internal rate of return)

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Deal too Good to be True?!…

- Land cost at historical value
- Mismatched growth rates between rental income and expenses
- Missing operating expenses
- Failure to consider rental concessions and effective rents

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Deal too Good to be True?!… *Continued*

- Vacancy rates too low
- Use of low going-out capitalization rates
- Underestimation of sales and other reversion costs

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Outlook

- Focus of next lecture
  - Risk assessment methods
  - Effects of financial leverage on return/risk
  - Concept of incremental cost of debt
  - Assess advantages and disadvantages of borrowing

…then development analysis and PE / funds

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